

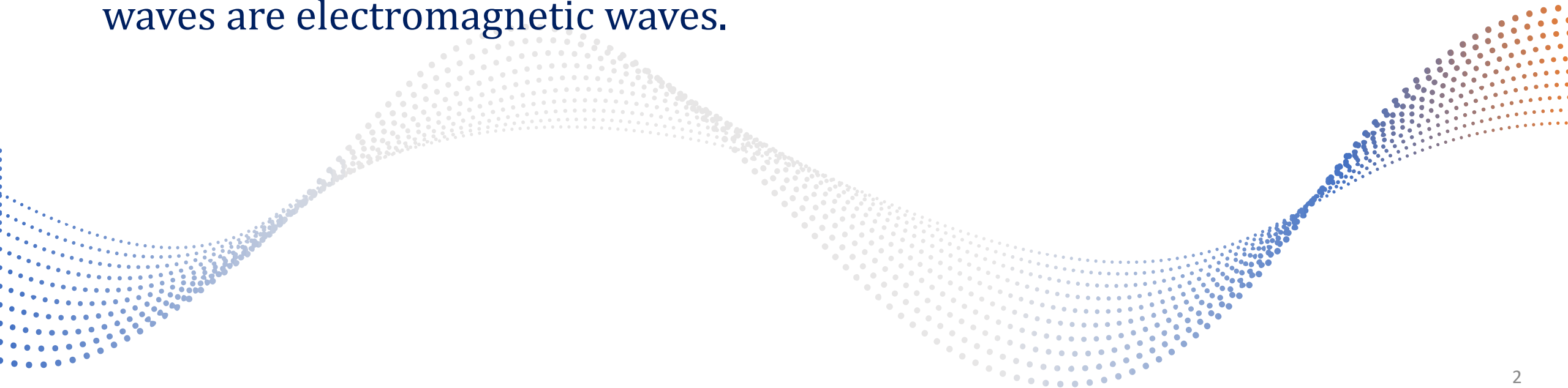
UNIT – I

Polarization

Module-1: Physical Optics: Polarization, Malus' Law, Brewster's Law, Double Refraction.

- **Introduction:**

- Interference and diffraction phenomena proved that light is a wave motion
- These phenomena are used to find wavelength of light and they do not give any indication regarding the character of waves.
- Maxwell developed electromagnetic theory and suggested that light-waves are electromagnetic waves.

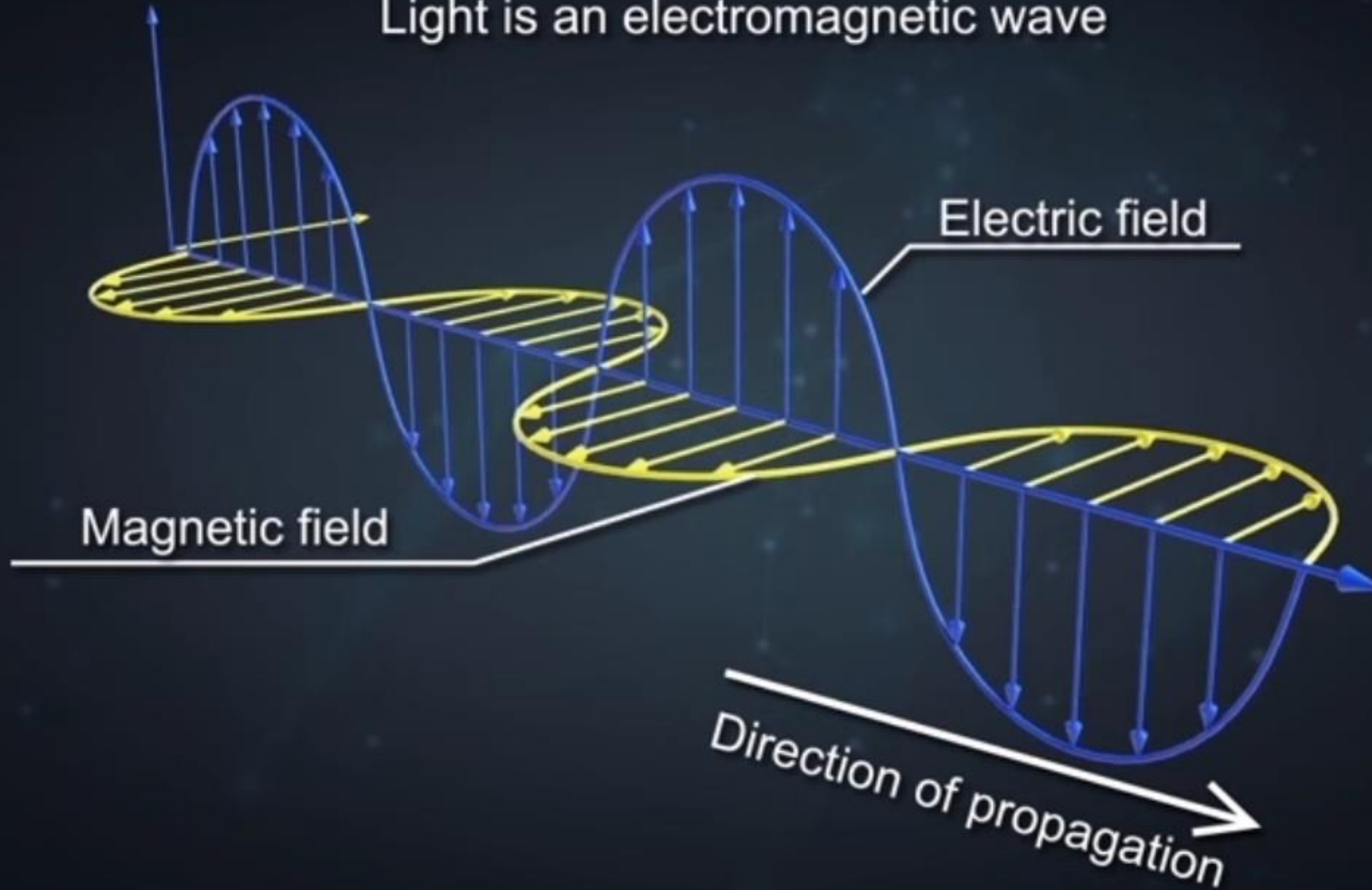


- ❖ Electromagnetic waves are transverse waves, so it is obvious that light waves are also transverse waves.
- ❖ Longitudinal waves are waves in which particles of medium oscillate along the direction of propagation of wave (e.g. sound wave).
- ❖ Transverse waves are waves in which particles of medium oscillate perpendicular to the direction of propagation of wave. (e.g. Electromagnetic waves.)
- ❖ Polarization is possible in transverse wave



POLARIZATION

Light is an electromagnetic wave



Mutually perpendicular

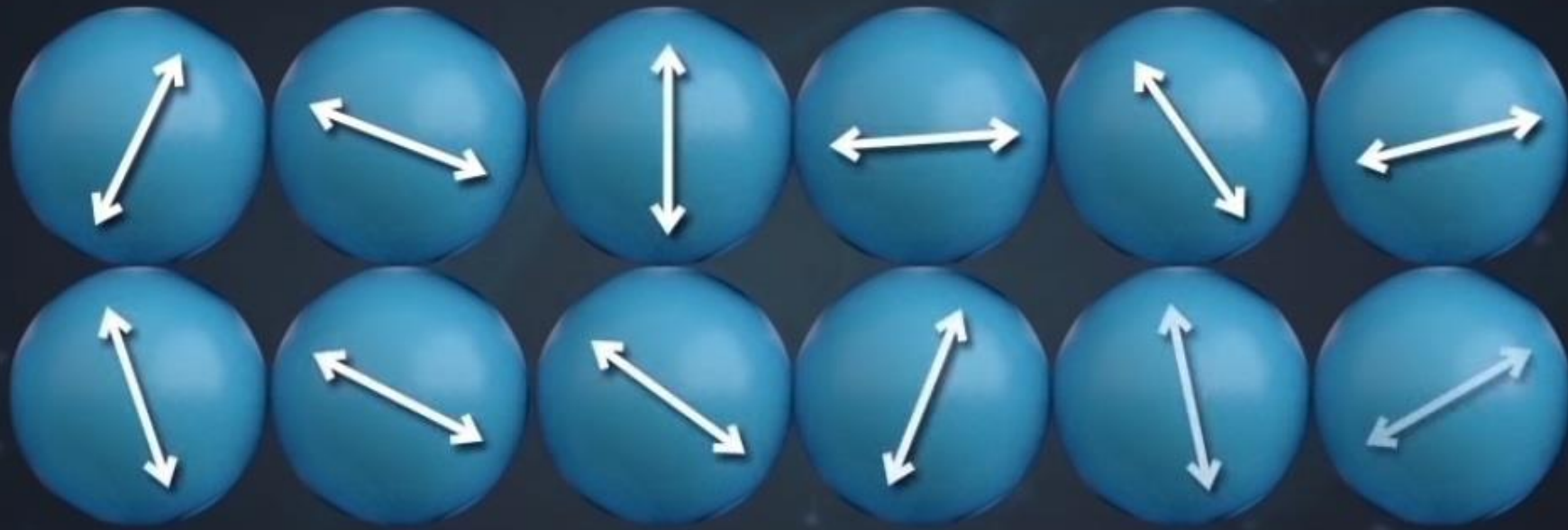
POLARIZATION



Electric field vibrations are responsible for the vision.
Vibration of light means electric field vibrations.

- **Unpolarized Light** is the light in which the planes of vibration are symmetrically distributed about the propagation direction of the wave.
- **Plane Polarized light** is a wave in which the electric vector is everywhere confined to a single plane.
- **A linearly polarized light** wave is a wave in which the electric vector oscillates in a given constant orientation.

POLARIZATION



POLARIZATION



Common source of light

Emit light that have electric field vibrations
in a number of directions.



Incandescent bulb



Candle flame



Florescent lamp





This type of light beam which has vibrations in more than one plane is called unpolarized light.



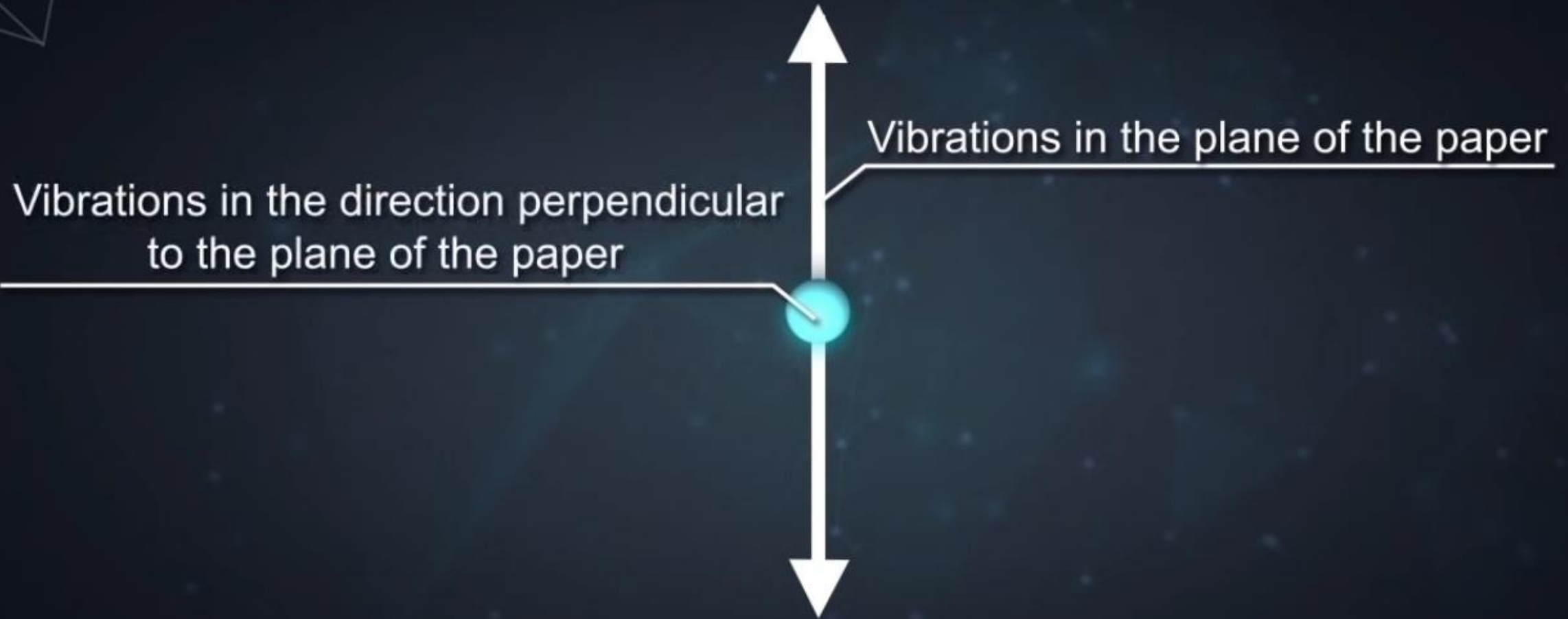
A light beam which has vibrations in only one plain is called polarized light.





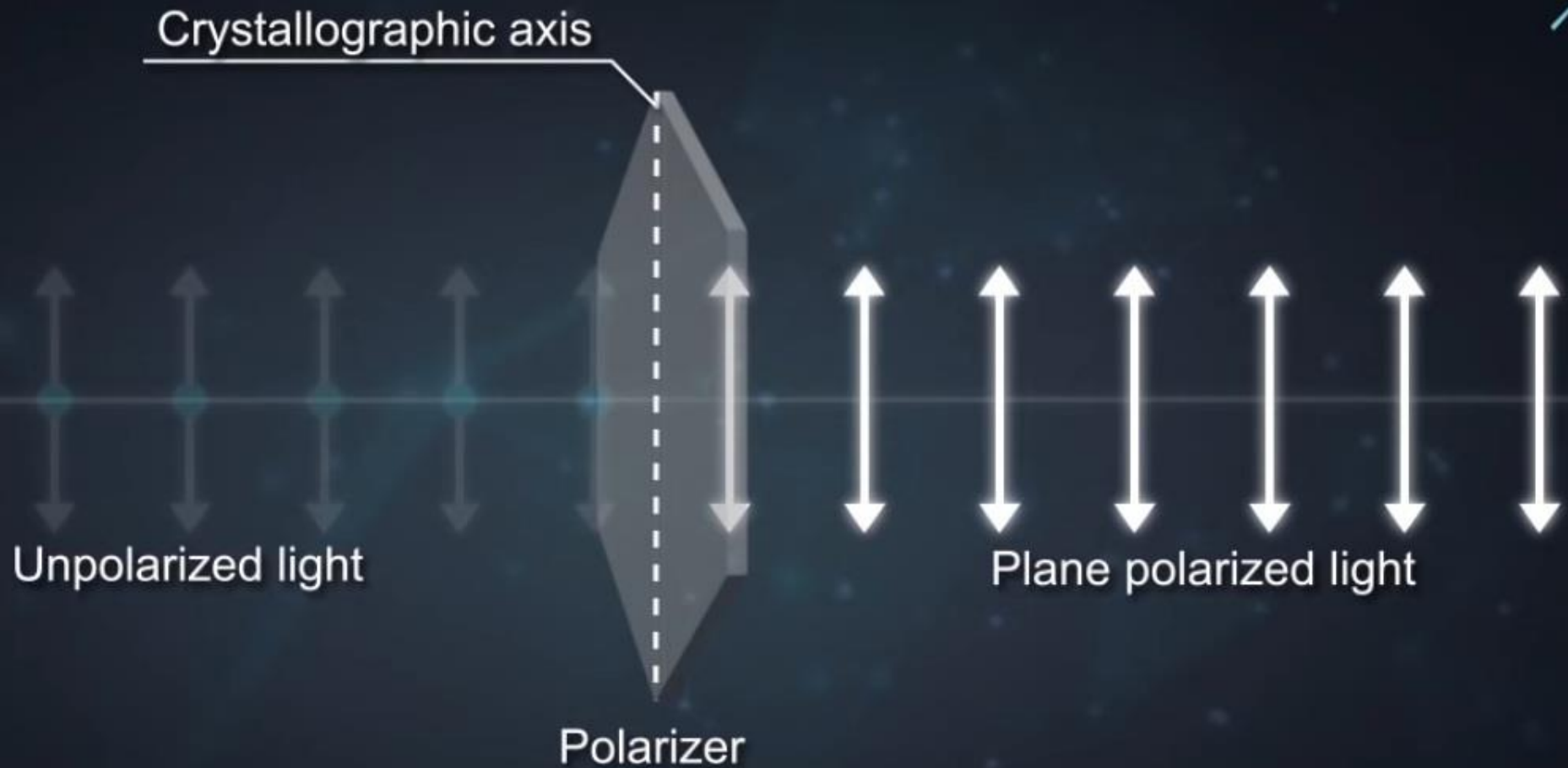
This is the simpler way of representing an unpolarized light.

POLARIZATION





POLARIZATION OF LIGHT



The phenomenon of restricting the vibrations of light to particular plane perpendicular to the propagation light is called polarization of light.

Polarizer and Analyser :

- **Polarizer:**

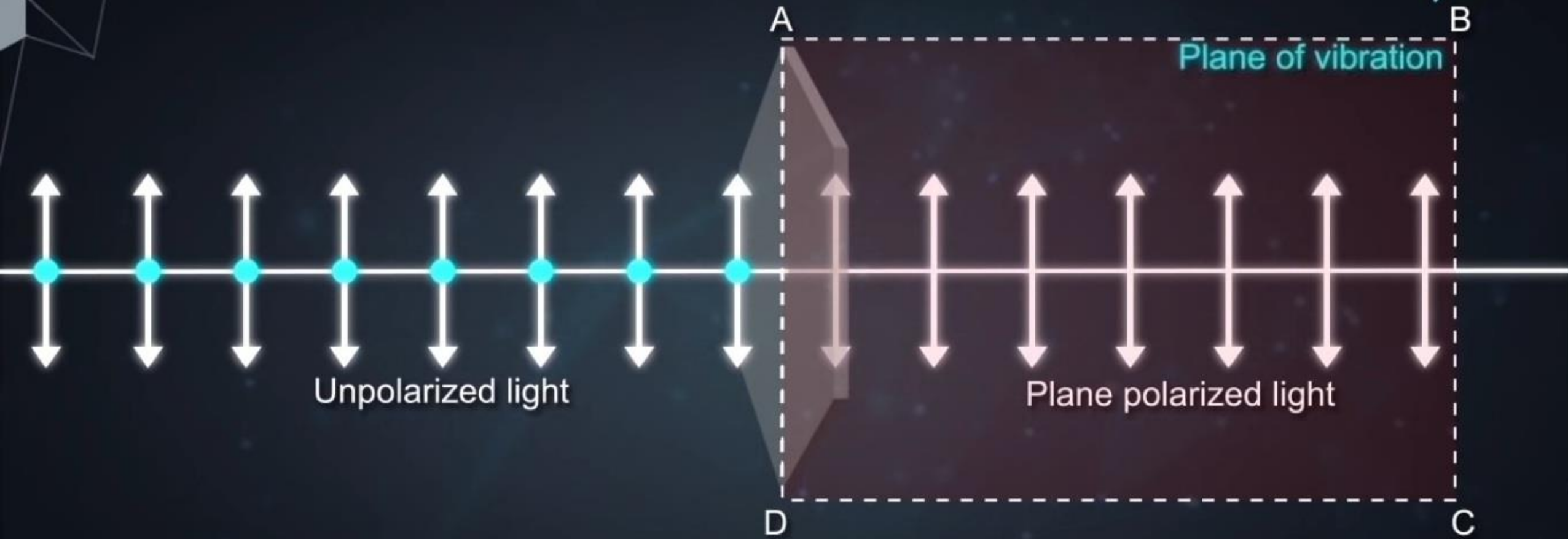
- A polarizer is device that transforms unpolarized light into polarized light.

- **Analyser:**

- An analyser is a device which is used to identify the direction of vibration of linearly polarized light.

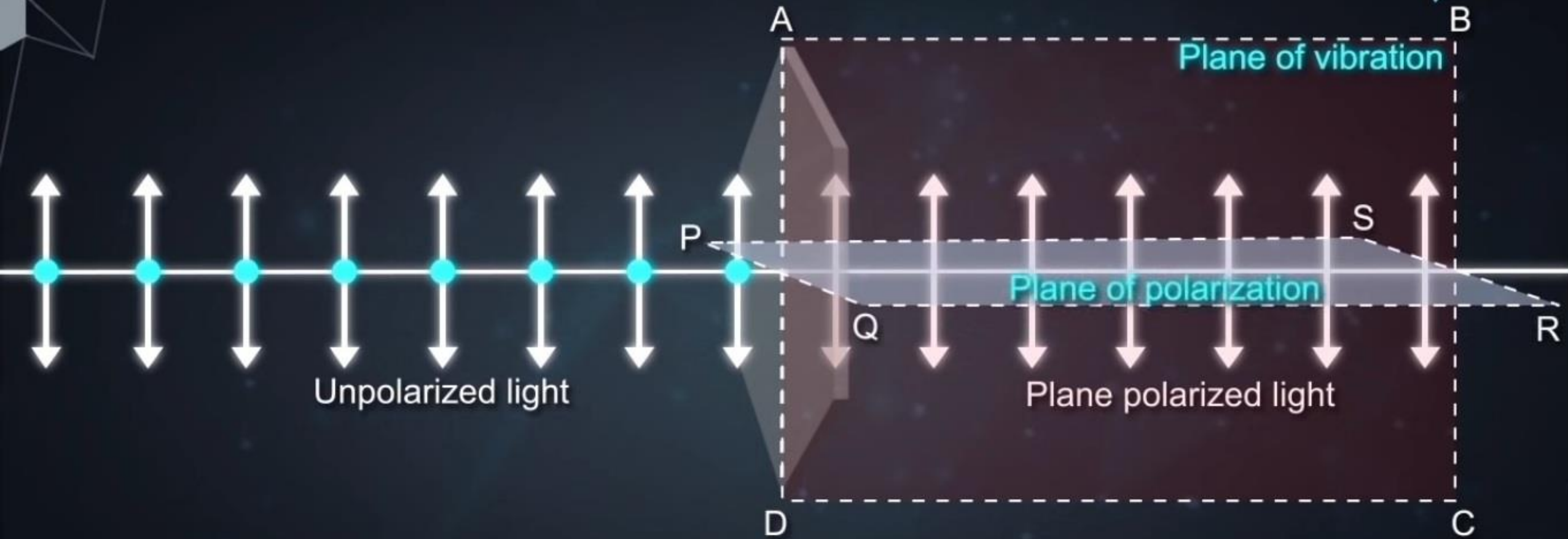
- Linear polarizers are prepared using birefringent crystals or dichroic crystals out of these Nicol crystal is widely used for linear polarizer.
- Polarizer sheets made using dichroism are known as polarized sheets.

POLARIZATION OF LIGHT



Plane of vibrations: The plane (ABCD) here that contains vibrations of plane polarized light is called plane of vibrations.

POLARIZATION OF LIGHT



Plane of polarization: The plane (PQRS) perpendicular to the plane of vibrations is called plane of polarization. It is also the plane in which there is no vibration.



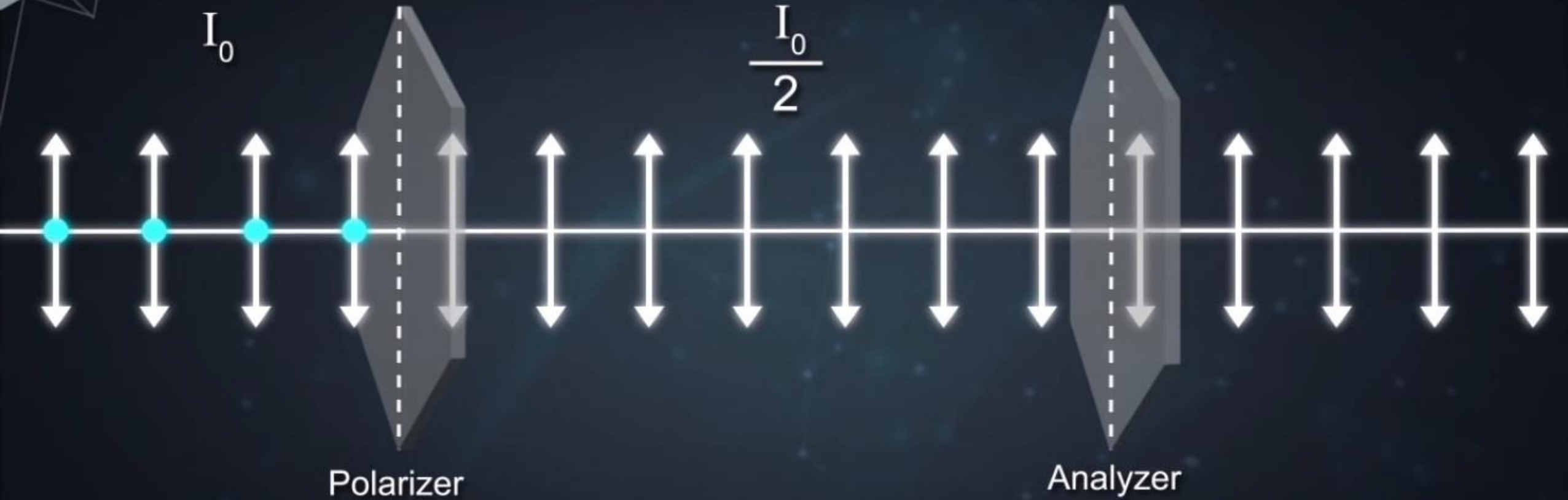
POLARIZATION OF LIGHT

Unpolarized light

$$I_0$$

Polarized light

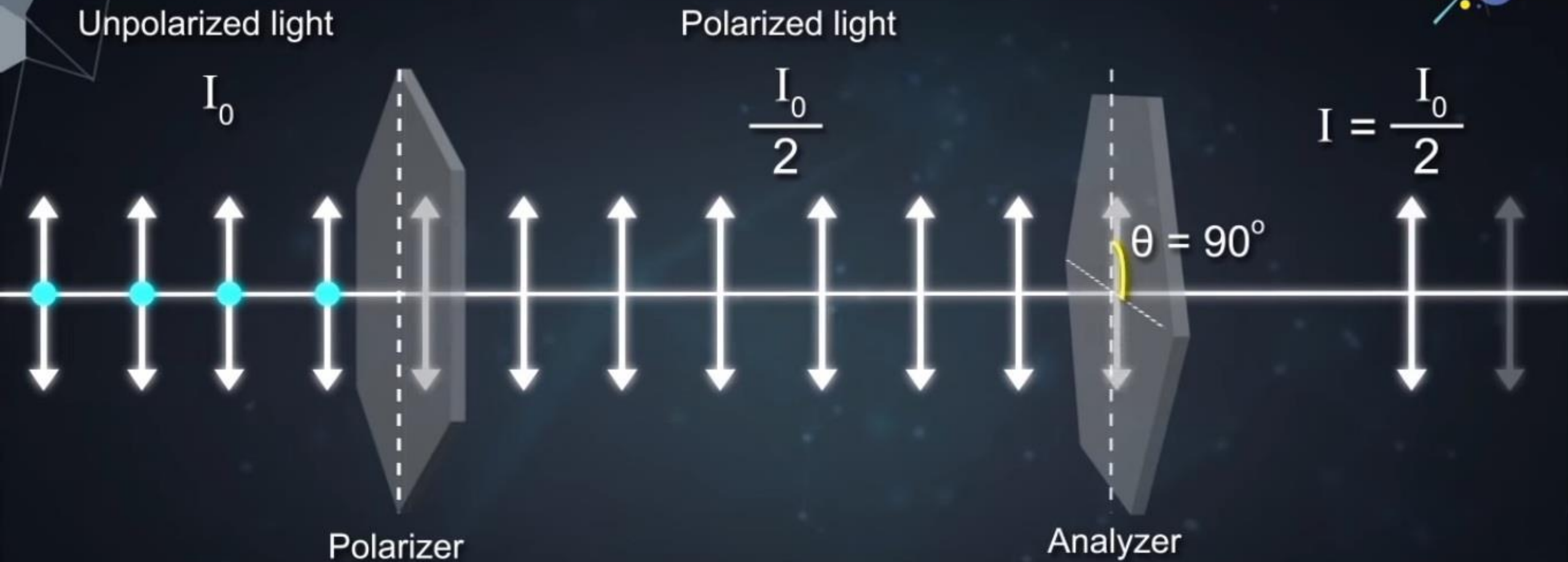
$$\frac{I_0}{2}$$



The emergent light will have no change in intensity, i.e. when $\theta = 0^\circ$ or 180° , the intensity is maximum, $I = I_0/2$.



POLARIZATION OF LIGHT



When the pass axis of the analyzer polarizer is perpendicular to the pass axis of the polarizer, no light will be transmitted out of the analyzer i.e. when $\theta = 90^\circ$, the intensity $I = 0$.

Production of Linearly Polarized Light:

➤ Methods for producing Linearly polarized light :

1. Reflection
2. Refraction
3. Scattering
4. Selective absorption
5. Double reflection.

➤ Applications of Polarized light:

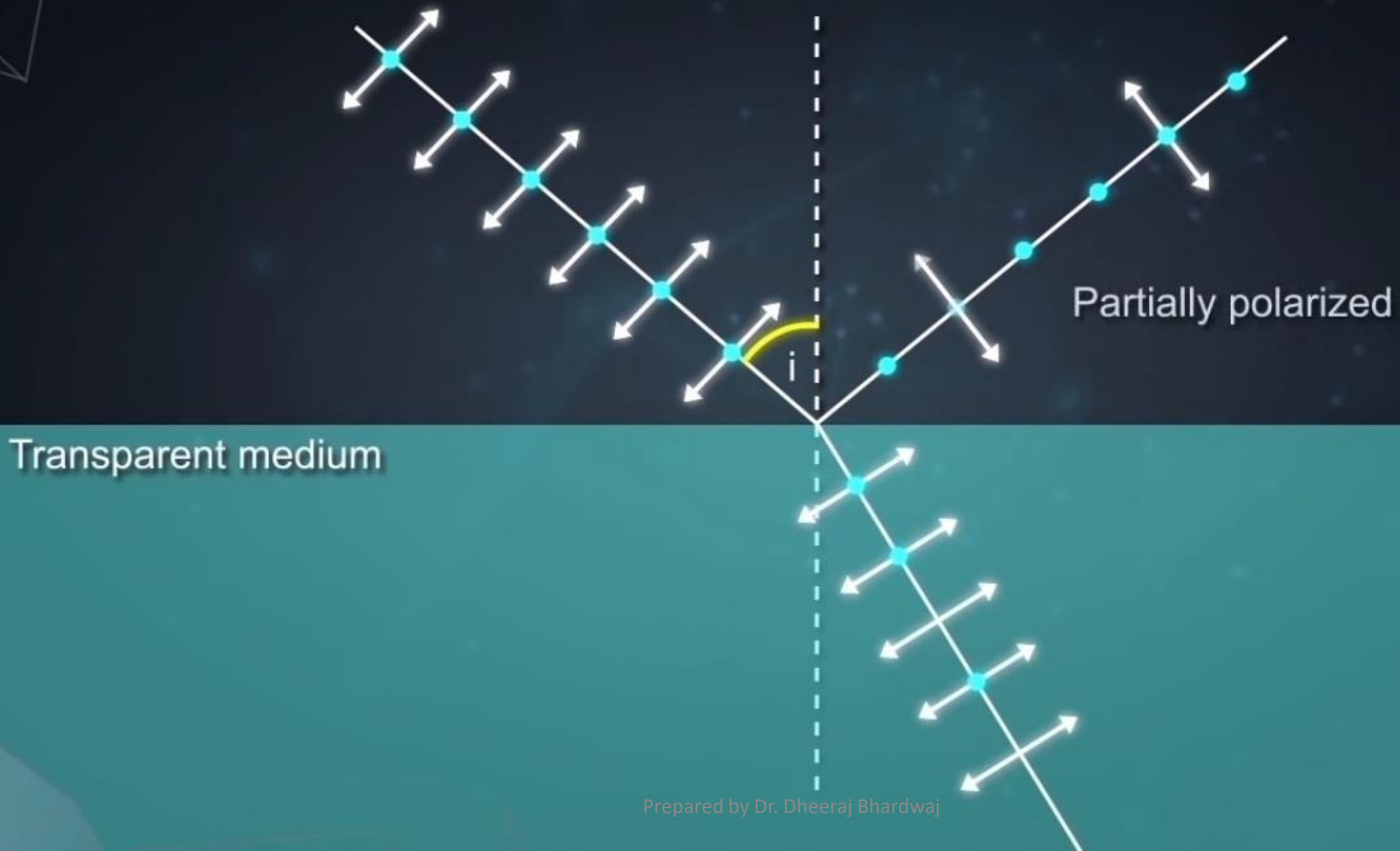
- I. Industry and Engineering fields
- II. In liquid crystal displays (LCDs)
- III. Widely used in wristwatches, calculators, T.V. Screens

Polarization by reflection:

- In 1808, E.L. Malus discovered the polarization of natural light by reflection from the surface of glass.
- He noticed that when natural light is incident on a plane sheet of glass at a certain angle, the reflected beam is plane polarized.

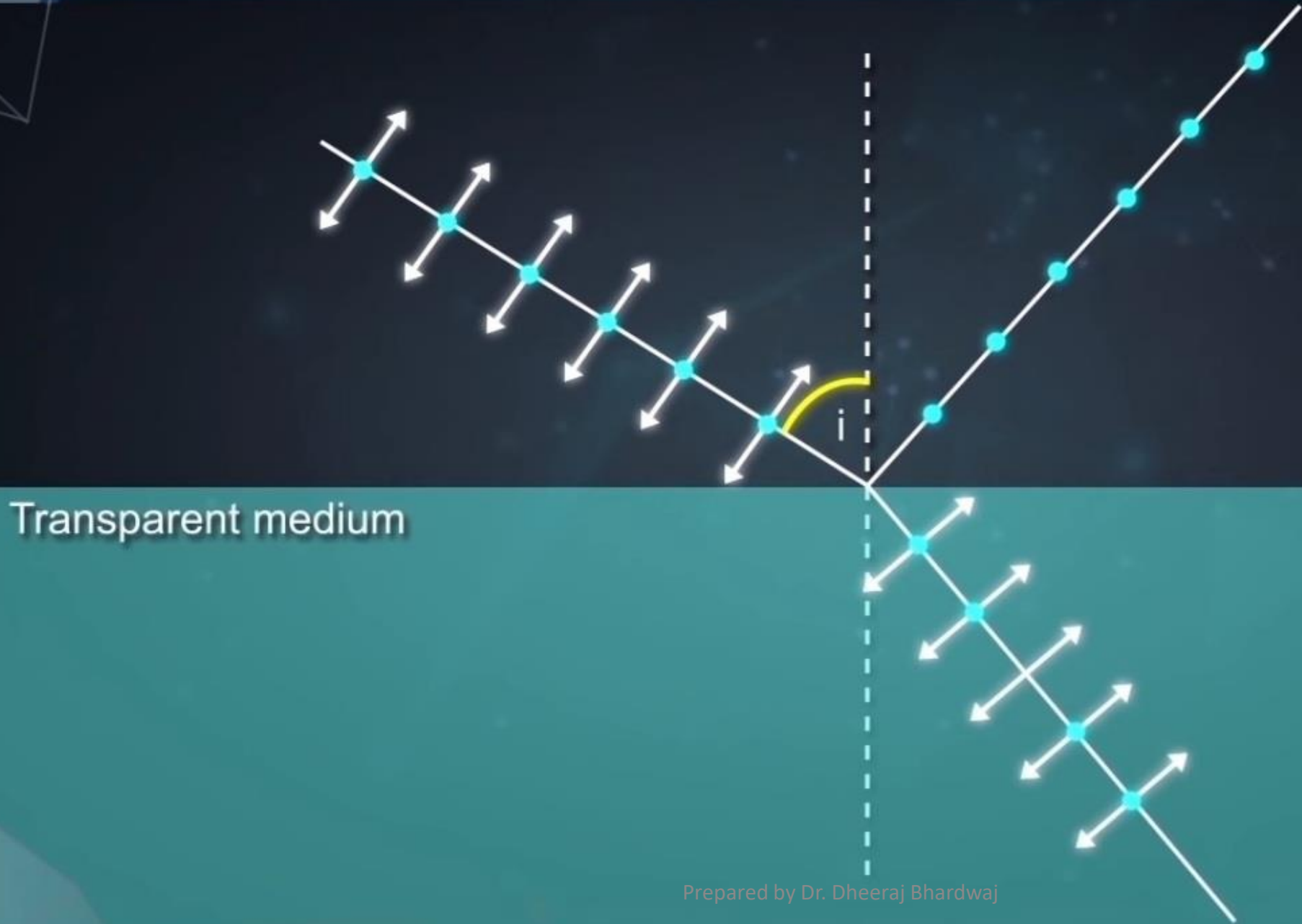


POLARIZATION BY REFLECTION



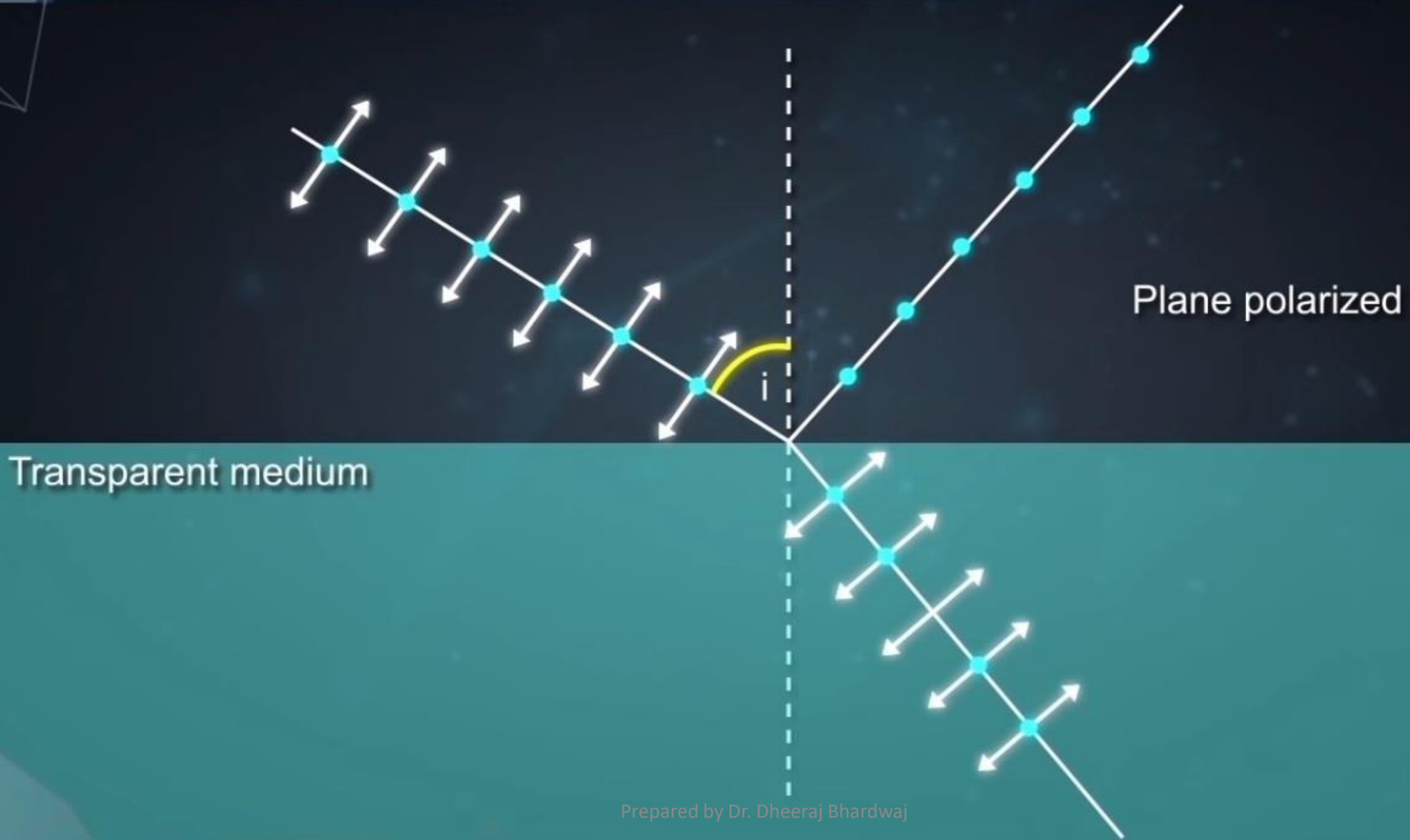


POLARIZATION BY REFLECTION





POLARIZATION BY REFLECTION





POLARIZATION BY REFLECTION



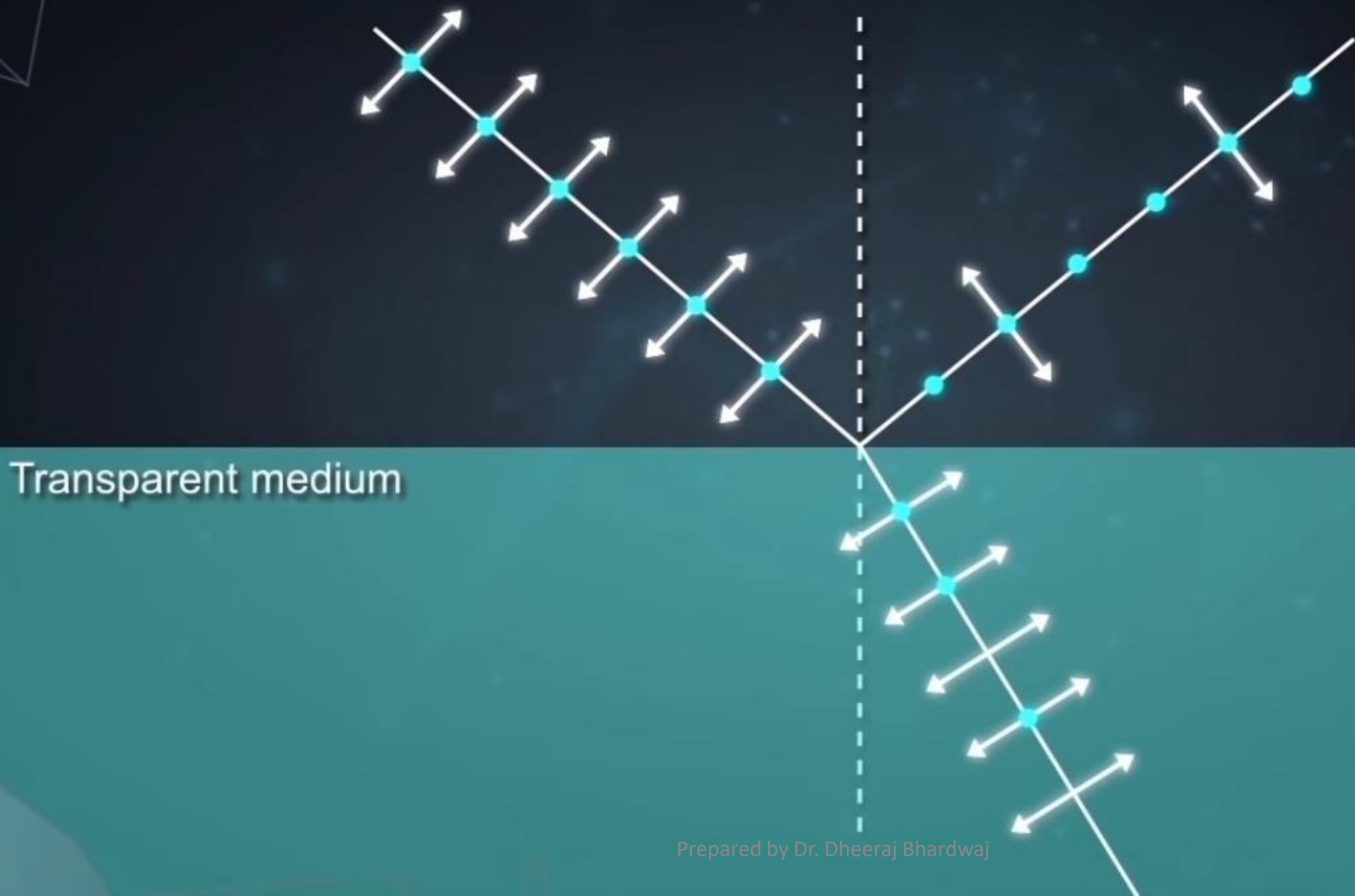
The simplest method to produce plane polarized light is by reflection



Malus
1808

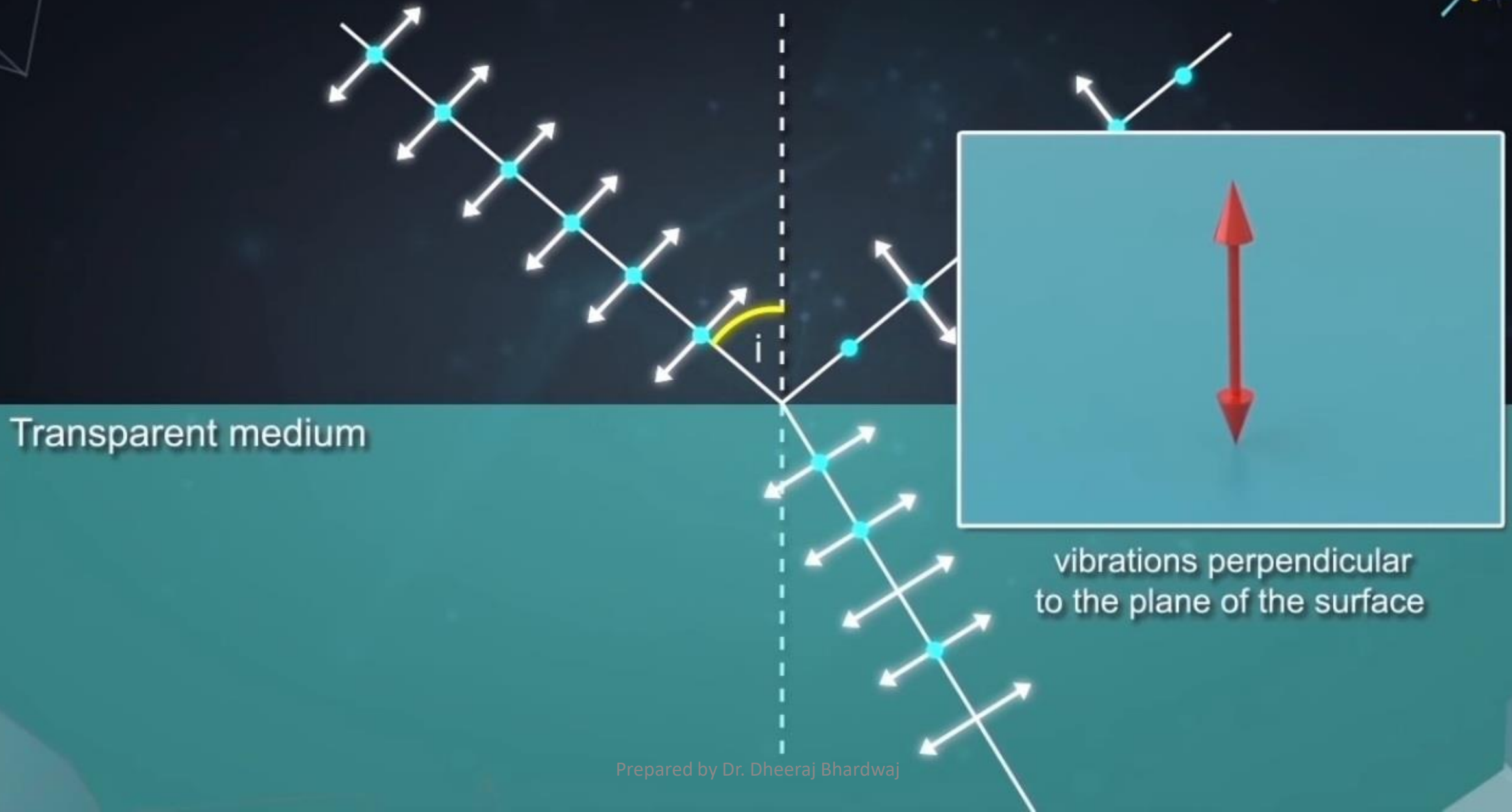


POLARIZATION BY REFLECTION



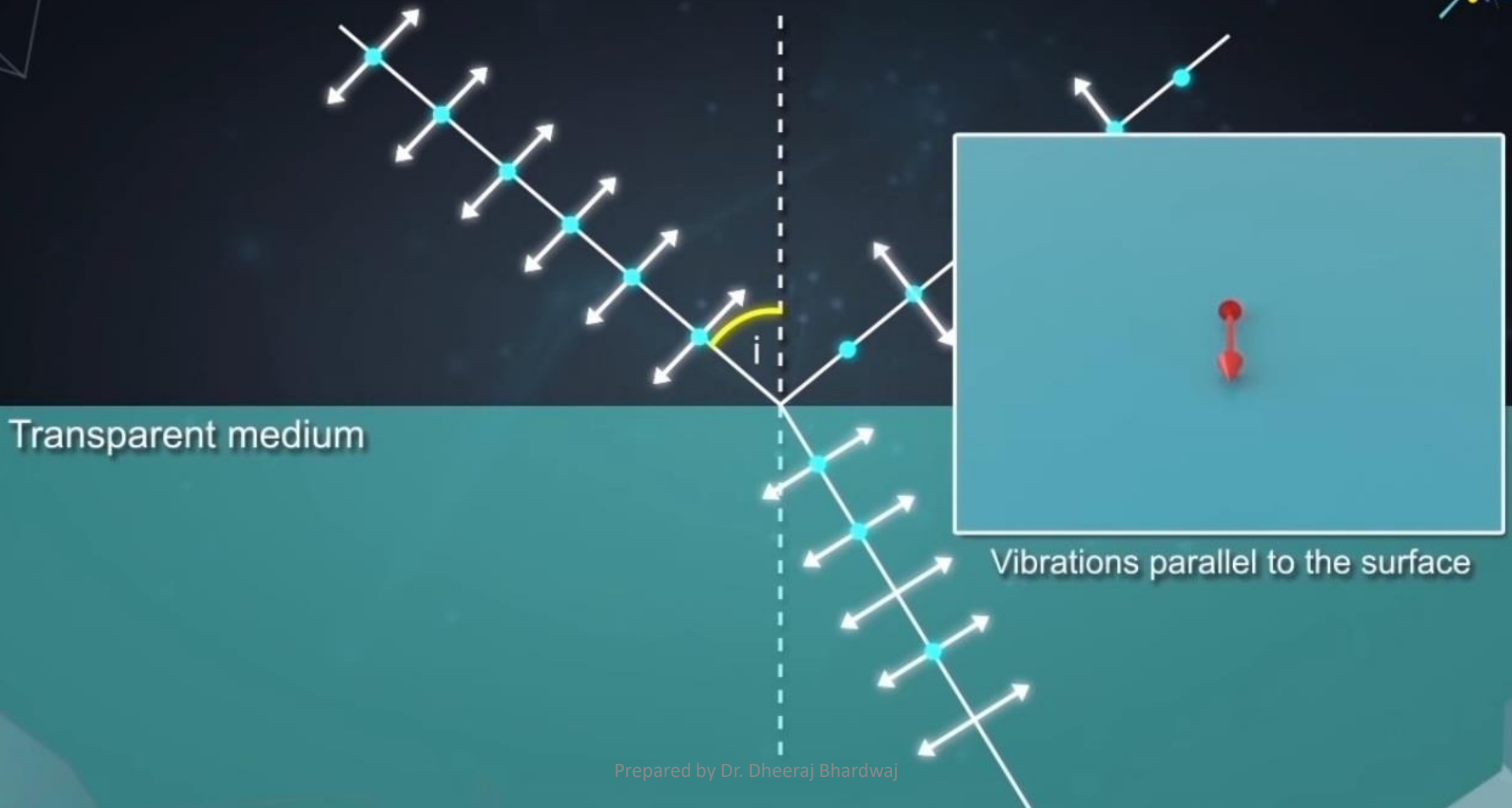


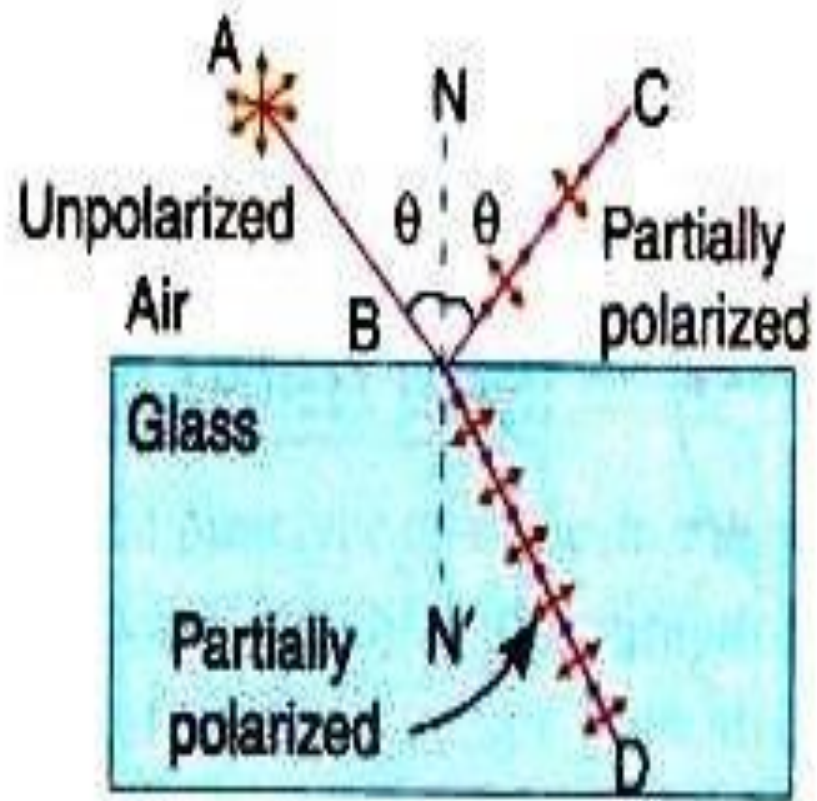
POLARIZATION BY REFLECTION



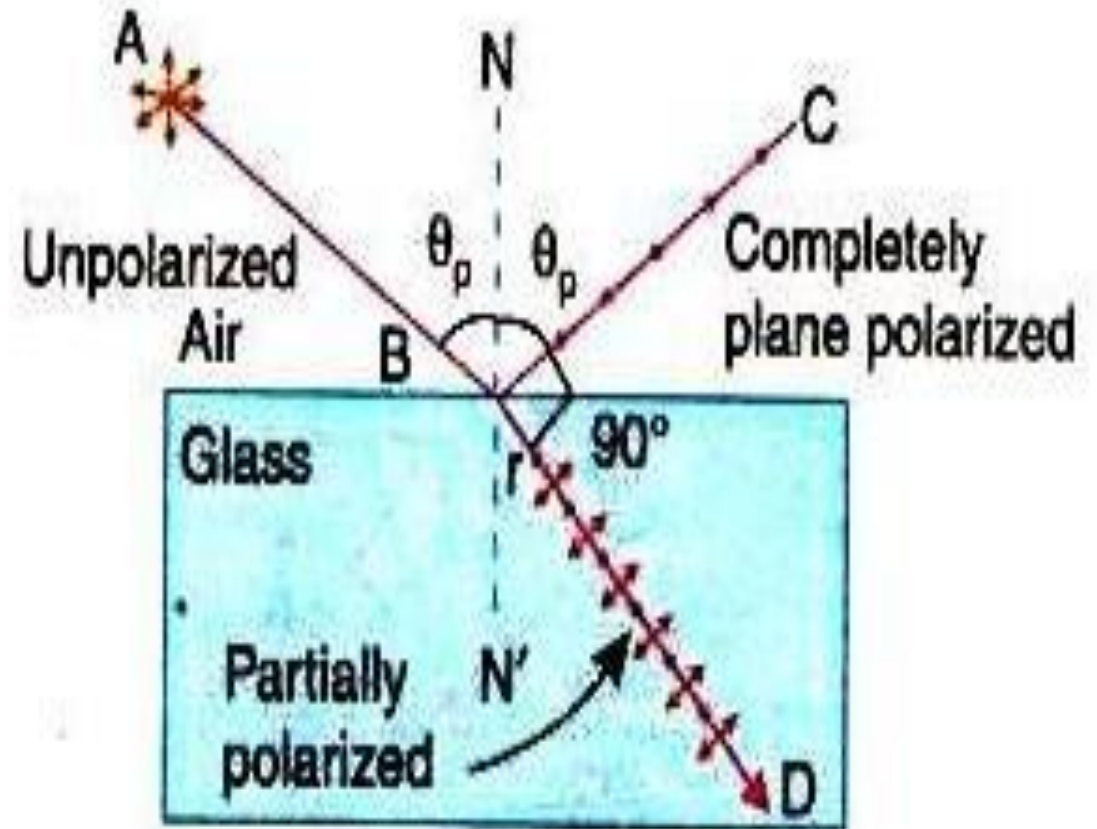


POLARIZATION BY REFLECTION





(a)



(b)



Malus law

When a completely plane polarized light is incident on an analyzer. The intensity of emergent light varies as the square of the cosine of the angle between the polarizer and analyzer.

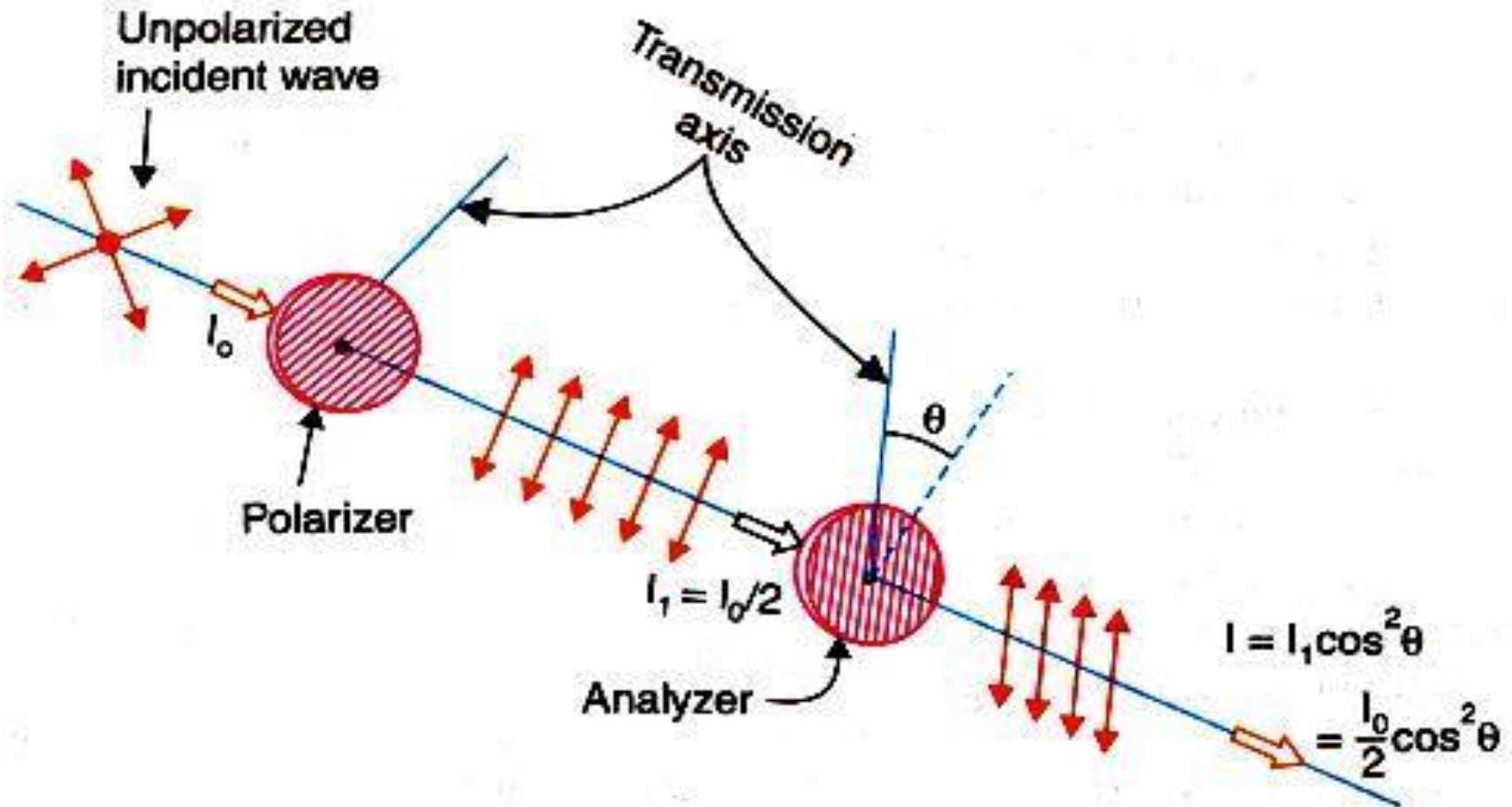
Effect of analyzer on plane polarized light- Malus law

When natural light is incident on a polarizer, the transmitted light is linearly polarized. If this light-further passes through an analyzer, the intensity varies with the angle between the transmissions axes of polarizer, and analyzer.

Malus studied their phenomenon and known as Malus law.

Statement:

"The intensity of the polarized light transmitted through the analyzer is proportional to cosine square of the angle between the plane of transmission of the analyzer and plane of transmission of the polarizer."



Let I_0 is the intensity of unpolarized light.

The intensity of polarized light from the polarizer is $I_0/2$.

Taking $I_1 = I_0/2$.

This plane polarized light then passes through the analyzer.

Let E is the amplitude of vibration and θ is the angle between this vibration and transmission axis of an analyzer.

E resolves into two components

1. E_y parallel to the plane of transmission of the plane of analyzer, and
2. E_x , perpendicular to the plane of analyzer.

E_y component is only transmitted through the analyzer.

Intensity of light for this component:

$$E_y = E \cos \theta$$

$$I = E^2 \cos^2 \theta = I_1 \cos^2 \theta = \left(\frac{I_0}{2} \right) \cos^2 \theta$$

If,

- i. $\theta = 0^\circ$, then axis are parallel $I = I_1$
- ii. $\theta = 90^\circ$, then axis are perpendicular $I = 0$
- iii. $\theta = 180^\circ$, then axis are parallel $I = I_1$
- iv. $\theta = 270^\circ$, then axis are perpendicular $I = 0$

Thus, there are two positions of maximum intensity and two positions of zero intensity when we rotate the axis of the analyzer with respect to that of the polarizer.

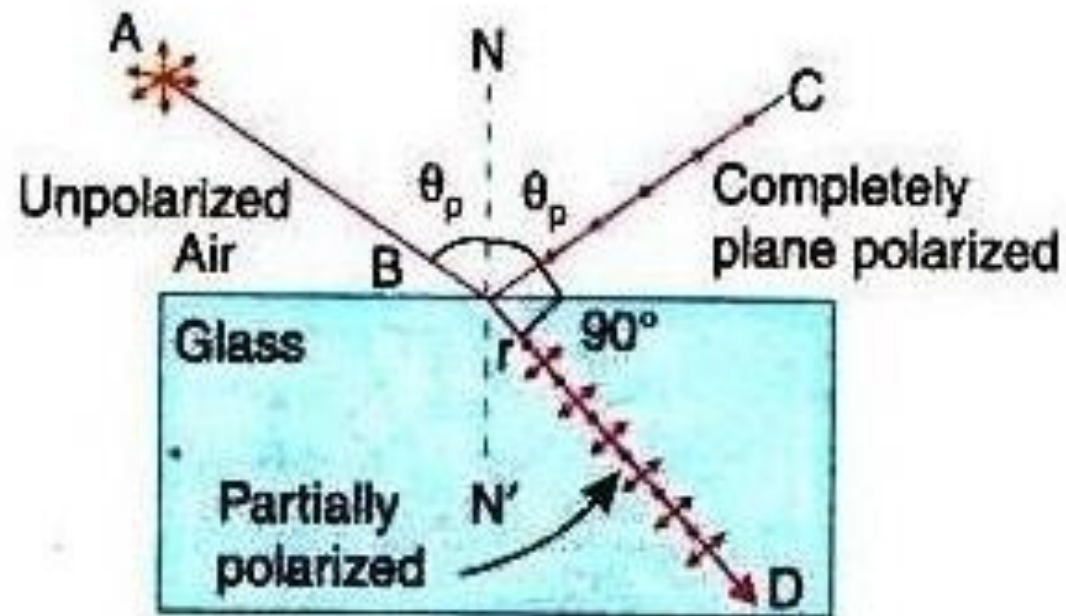
Prove that the reflected and the refracted rays are at right angles to each other in case of the Polarization phenomenon (Slides 38 to 42)

Brewster's Law

- In 1892, Brewster performed number of experiments to study the polarization of light by reflection at the surfaces of different media.
- He found that ordinary light is completely polarized in the plane of incidence when it gets reflected from a transparent medium at a particular angle known as the '*angle of polarization.*'
- He proved that '*the tangent of the angle of polarization is numerically equal to the refractive index of the medium.*'
- Also the reflected and retracted rays are perpendicular to each other.

If θ_P is the angle and μ is the refractive index of the medium, then This is known as Brewster's law.

$$\mu = \tan \theta_p$$



Brewster found that the maximum polarization of reflected ray occurs when it is at right angles to the refracted ray. It means that

According to Snell's law,

Where, μ_2 is the absolute refractive index of reflecting surface
 μ_1 is the refractive index of the surrounding medium.

$$\theta_p + r = 90^\circ$$

$$\frac{\sin \theta_p}{\sin r} = \frac{\mu_2}{\mu_1}$$

$$\frac{\sin\theta_p}{\sin(90^\circ - \theta_p)} = \frac{\mu_2}{\mu_1}$$

$$\frac{\sin\theta_p}{\cos\theta_p} = \frac{\mu_2}{\mu_1}$$

$$\tan\theta_p = \frac{\mu_2}{\mu_1}$$

The polarizing angle θ_p is also known as **Brewster angle** and denoted by θ_B

From Snell's law, $\mu = \frac{\sin i}{\sin r}$

From Brewster's law, $\mu = \tan i = \frac{\sin i}{\cos r}$

Comparing above equations,

$$\cos i = \sin r = \cos (\pi/2 - r)$$

$$i = \pi/2 - r$$

$$i + r = \pi/2$$

Therefore the reflected and the refracted rays are at right angles to each other.

From Brewster's law, the value of 'i' for crown glass of refractive index 1.52 is given by.

$$i = \tan^{-1}(1.52)$$
$$i = 56.7^\circ$$

For ordinary glass the approximate value for the polarizing angle is 57° .

For a refractive index of 1.7, the polarizing angle is 59.5° . Thus the polarizing angle is not widely different for different glasses.

Applications of Brewster's law

1. Brewster's law can be used to determine the reflective indices of opaque materials.
2. It is used to calculate the polarizing angle for total polarization of reflected light, if reflective index of the material is known.
3. Brewster's angle can be utilized for transmitting a light beam in into or out of an optical fiber without reflections losses.

Anisotropic Crystals:

➤ Isotropic Materials:-

In isotropic materials, atoms are arranged in a regular periodic manner. In isotropic materials, when a light beam is incident, it refracts a single ray. It means that in such material the refractive index is same in all direction. e. g. Glass ,water and air

➤ Anisotropic Materials:-

In anisotropic material, the arrangement of atoms differs in different directions within a crystal. Thus the physical properties vary like, thermal conductivity, electrical conductivity, velocity of light and refractive index etc. vary with the directions. Such crystal are then said Anisotropic.

The anisotropic crystals are divided into two classes.

(i) Uniaxial Crystal: In this type of crystal, one of the refracted rays is an ordinary ray and the other is an extraordinary. e. g. Calcite, tourmaline and Quartz.

(ii) Biaxial Crystal: In biaxial crystal both the refracted rays are extraordinary rays. e. g. mica, topaz & aragonite.

Calcite Crystal:

Calcite or Iceland spar is crystallized calcium carbonate (CaCO_3), it is very transparent crystal.

Calcite crystal is the form of a rhombohedron bounded by six parallelograms with angles equal to 78 and 102.

At two opposite corners (A&H) the three angles of faces meeting there are all obtuse (larger than 90°)

These corners (A&H) are known as blunt corners.

➤ Optic axis

A line passing through 'A' making equal angles with each of the three corners gives the direction of optic axis. Any line parallel to this line is also an optic axis.

Optic Axis:

At two opposite corners A and H of the rhombohedron all the angle of faces are obtuse (Fig 1).

The corners A and H are known as blunt corners.

A line drawn through A making equal angles with each of three edges gives the direction of the optic axis.

In fact any line parallel to this line is known as optic axis.

Optic axis is not a line but it is a direction.

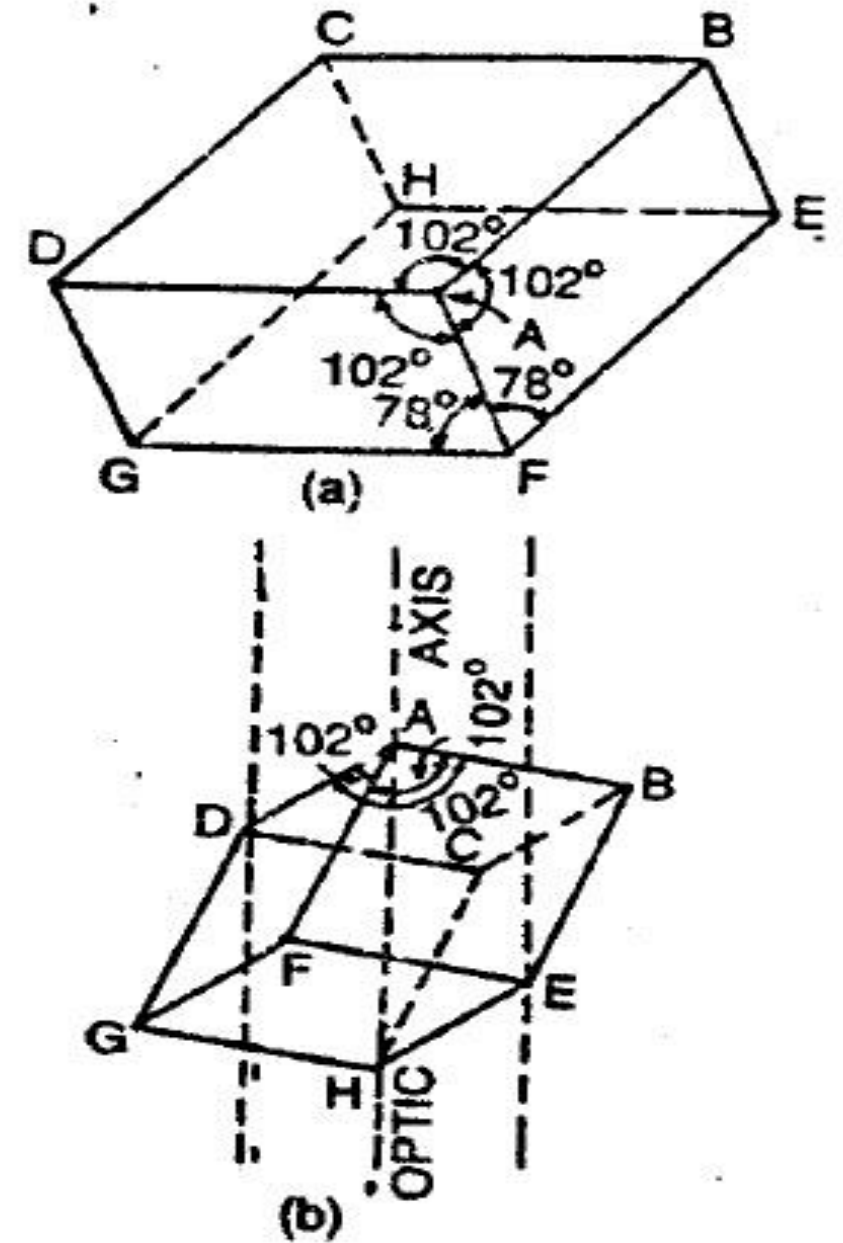
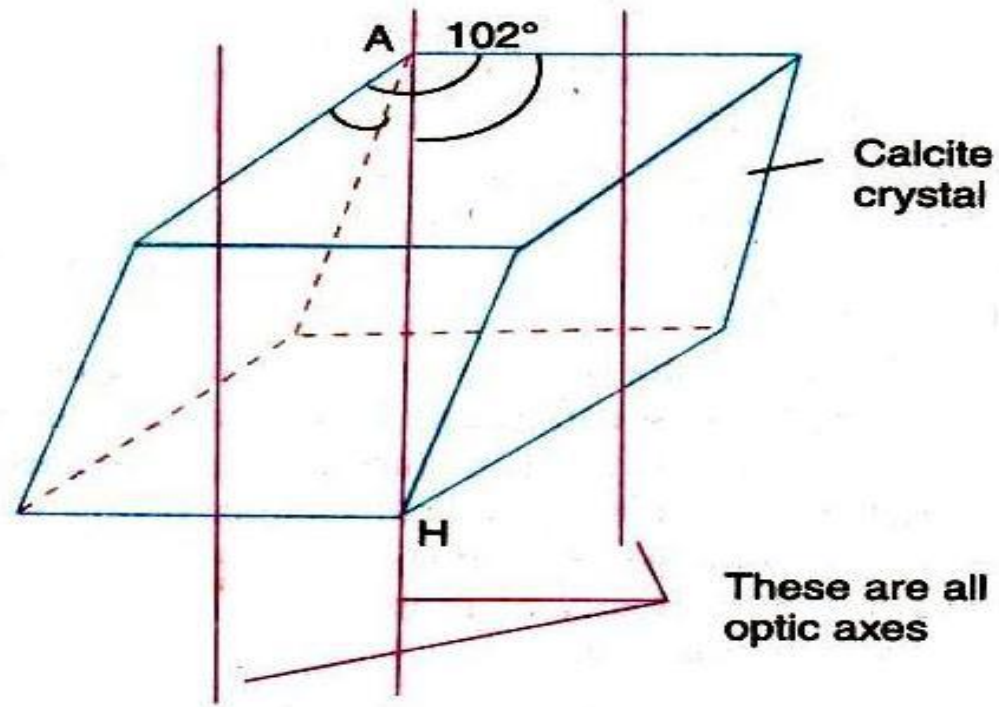


Fig. 1



AH is the optic axis of calcite crystal.

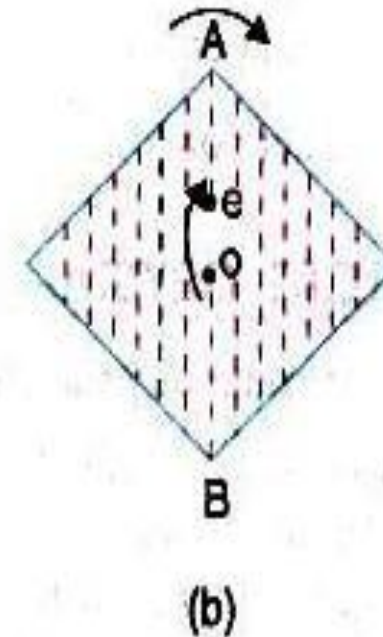
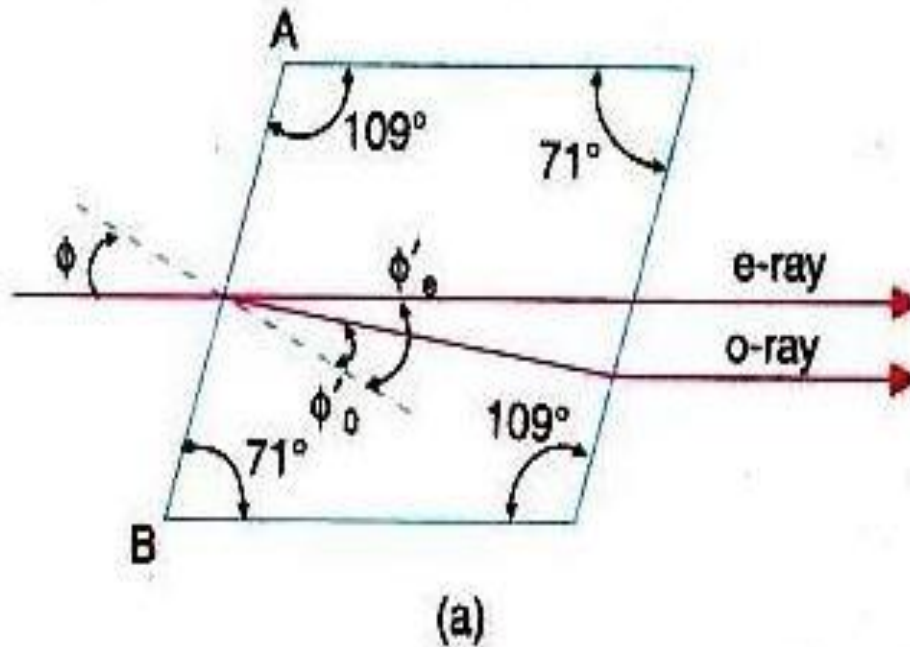
If a ray of light is incident along the optic axis or in a direction parallel to the optic axis, then it will not split up into two rays.

Thus the phenomenon of double refraction is absent when the light is allowed to enter the crystal along the optic axis.

Principal Section

- ❖ A plane containing the optic axis and perpendicular to a pair of opposite faces of the crystal is called the principal section of the crystal for that pair of faces.
- ❖ As a crystal has six faces, so far every point inside the crystal there are three principal sections, one for each pair of opposite crystal faces.

A principal section of a crystal is shown in the figure. The angles 71° and 109°



In fig(a) the principal section of the crystal is shown.

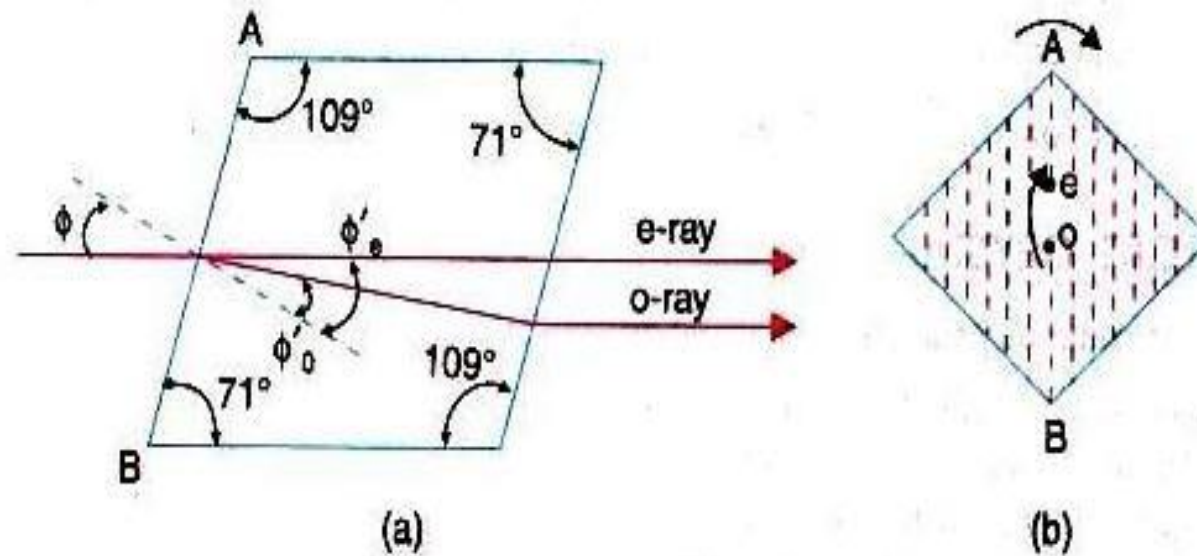
An end view of any principal section is a straight line (shown by dotted line in fig.b)

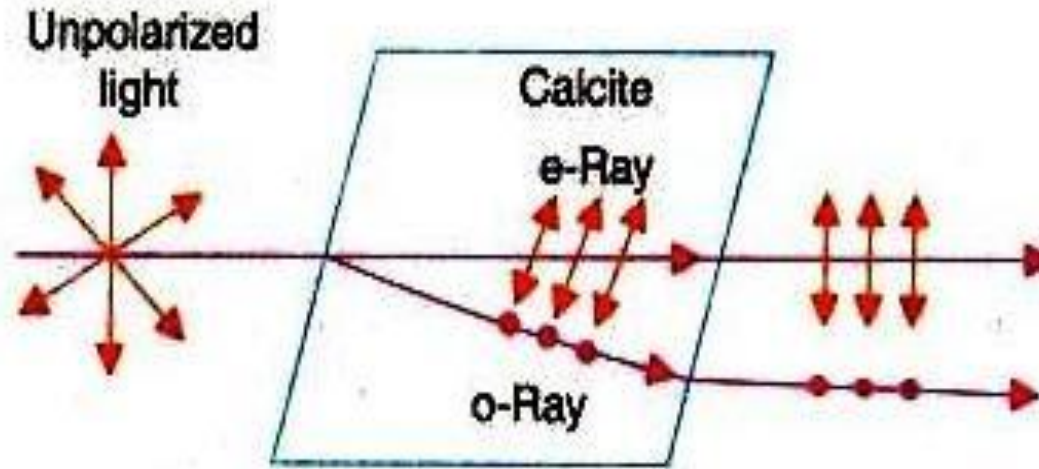
The plane containing the optical axis and the O -ray is called the principal plane of O -ray.

The plane containing the optic axis and the E-ray is called the principal plane of E-ray.

Double refraction

When a ray of light is refracted by a crystal of calcite it gives two refracted rays. This phenomenon is called "**DOUBLE REFRACTION**".

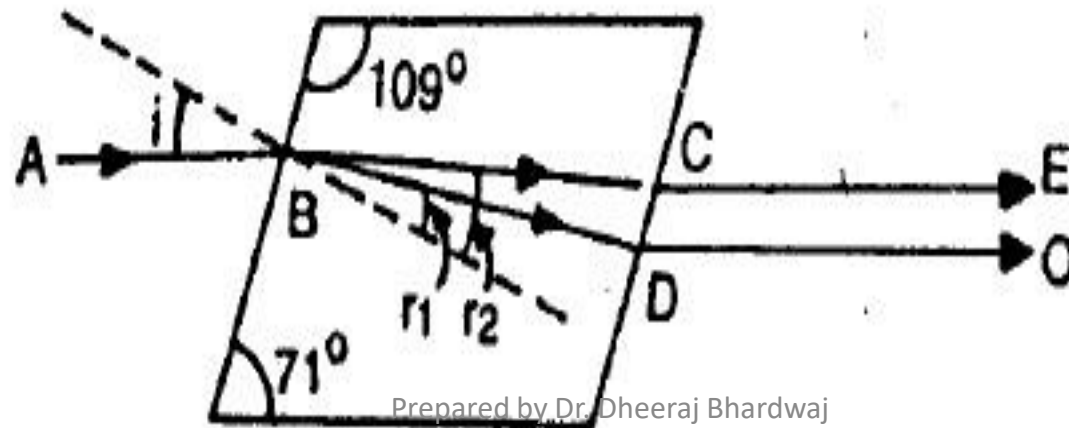




- The two rays produced in double refraction are linearly polarized in mutually perpendicular directions.
- The ray which obeys Snell's law of refraction is known as **ordinary ray** or **o-ray**.
- The other ray does not obey Snell's law is called **extraordinary ray** or **e-ray**.

Polarization by double refraction

- This phenomenon was discovered by **Erasmus Bartholinus** in 1669.
- When light is incident on a calcite crystal, it splits into two refracted rays.
- This phenomenon is called **double refraction** or **birefringence**. The crystal is called **birefringent**.



➤ **Positive crystal:**

When refractive index for extraordinary ray is greater than that of O-ray $\mu_e > \mu_o$.

➤ **Negative crystal:**

when refractive index for extraordinary ray is lesser than that of O-ray $\mu_e < \mu_o$

Mark an ink dot on a piece of paper.

If we place a calcite crystal over this dot, then two images of dots are observed.

Now rotate the crystal slowly as shown in figure ii.

It is found that one image remains stationary and the second image rotates with the rotation of the crystal.

The stationary image is known as the ordinary image

The second image is known as the extraordinary image.

The refracted ray which produces ordinary image is known as **ordinary ray O-ray**

The refracted ray which produces extraordinary image is known **extraordinary ray (E-ray)**.

When a ray of light AB is incident in the calcite crystal making an angle of incidence i , it is refracted along two paths inside the crystal

1. Along BC making an angle of refraction r_2 and
2. Along BD making an angle of refraction r_1 .

These two rays emerge out along CE and DO are parallel.

The difference between o-ray and e-ray is given below:

The ordinary ray has a refractive index $\mu_0 = \frac{\sin i}{\sin r_1}$

The extraordinary ray has a refractive index $\mu_e = \frac{\sin i}{\sin r_2}$

The o-ray obeys the laws of refraction and its refractive index is constant.

For e-ray its refractive index varies with the angle of incidence and it is not fixed.

For the case of calcite $\mu_o > \mu_e$ because r_1 less than r_2 .

Therefore, the velocity of light for the o-ray inside the crystal is less than the velocity of light for e-ray.

$$\mu_o = \frac{c}{v_o} \quad \text{and} \quad \mu_e = \frac{c}{v_e}$$

The o-ray travels in the crystal with same velocity in all directions

The velocity of e-ray is different in different directions, because its refractive index varies.

Both o-ray and e-ray is plane polarized.

They are polarized in mutually perpendicular planes.

Polarizer and Analyzer

➤Polarizer:-

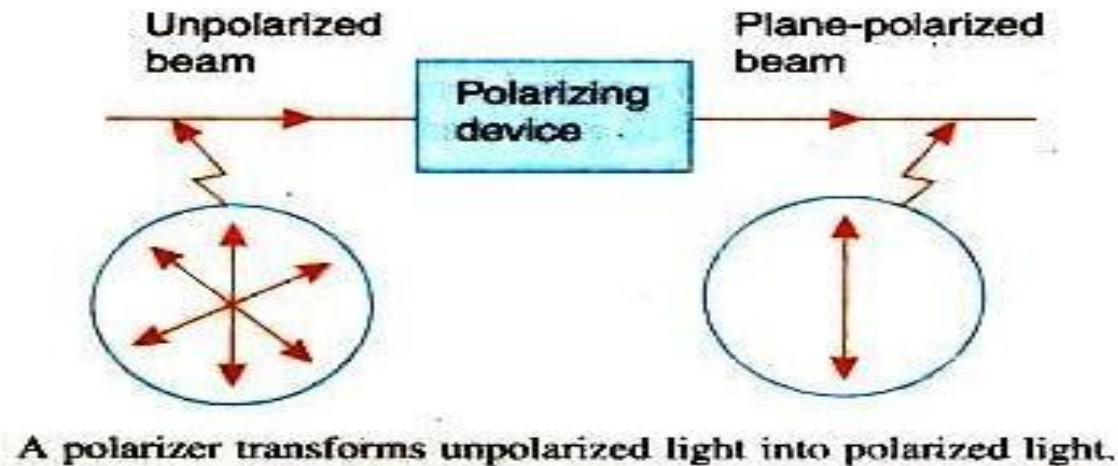
It is an optical device that transforms unpolarized light into polarized light. If it produces linearly polarized light. It is called a **lineally Polarizer**.

If natural light is incident on a linear polarizer, only that vibration which is parallel to the transmission axis is allowed to pass through the polarizer while the vibration that is in a perpendicular direction is totally blocked.

➤ Analyzer:-

Analyzer is a device, which is used to find whether the light is polarized or unpolarized.

Both polarizer and analyzer are fabricated in the same way and have the same effect on the incident light.



Uses of Polaroid

1. Polaroids are used in the laboratory to produce and analyse plane polarised light.
2. Polaroids are widely used as polarising sun glasses.
3. They are used to eliminate the head light glare in motor cars.
4. They are used to improve colour contrasts in old oil paintings.
5. Polaroid films are used to produce three - dimensional moving pictures.
6. They are used as glass windows in trains and aeroplanes to control the intensity of light. In aero plane one polaroid is fixed outside the window while the other is fitted inside which can be rotated. The intensity of light can be adjusted by rotating the inner polaroid.
7. Aerial pictures may be taken from slightly different angles and when viewed through polaroids give a better perception of depth.
8. In calculators and watches, letters and numbers are formed by liquid crystal display (LCD) through polarisation of light.
9. Polarisation is also used to study size and shape of molecules.





Photo taken without a polarised filter - reflected sunlight 'washes out the colour'. So much 'white light' has been reflected that the colour of the refracted light is less intense.



Photo taken with a polarised filter over the lens - reflected light has been reduced and there is a greater intensity of the refracted light from below the water surface.

Polarization In Everyday Situations



This photograph should be compared with the one above. In this photo, the reflected glare that was seen on the water's surface (previous photo) has been removed by the use of a Polaroid filter. It is much easier to see the sidewalk below the water surface.